

Contrast-based artifact removal enables microstate analysis in ambulatory EEG

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Abstract— Objective: Recent advances in electroencephalography (EEG) technology present new opportunities for mobile neuroimaging and real-world human neuroscience studies. However, EEG is sensitive to many sources of artifacts, and it can be especially difficult to remove high amplitude motion artifacts. Methods: We demonstrate a novel method using generalized eigen decomposition (GED) for artifact removal, validated this approach using semi-simulated and real artifactual EEG collected during walking and jogging, and showed the feasibility of using cleaned ambulatory EEG data for brain microstate analysis. Results: We found that GED is effective even in ultra-low SNR (0.1 – 5) conditions, achieving a correlation of 0.93 and RMSE of $1.43 \mu V$ in recovering ground truth activity using semi-simulated data, and increased the number of brain components by 10.9 and 11.8 for real data. GED showed superior performance compared with artifact subspace reconstruction (ASR) and independent component analysis (ICA) methods on semi-simulated data in very low SNR regimes. Using cleaned data, we were able to extract canonical EEG microstates across all tasks and sessions for examining task-related modulation in microstate duration, occurrence, and time coverage. We observed increased duration, occurrence and time coverage of microstates A and duration of microstate B, and decreased occurrence and time coverage of microstate D during motion compared with rest, corresponding to heightened alertness and increased visual processing. Conclusion: These findings not only validate our artifact removal approach but also open new avenues for investigating neural dynamics during naturalistic human behavior.

Index Terms— electroencephalography, motion artifacts, real-world neuroimaging, generalized eigen decomposition, microstate analysis, ecological validity.

I. INTRODUCTION

THE human brain orchestrates a vast range of complex behaviors. Despite tremendous advancements in human neuroscience, studying the brain in real-world

settings remain a significant challenge due to technological and methodological constraints [1][2]. Traditional studies mostly rely on controlled and stationary laboratory experiments [3] leading to a scarcity of diverse longitudinal brain data in naturalistic settings and limiting insight into brain function that underlie the rich and complex behaviors experienced in the real world. While attempts have been made to address this gap by simulating real-life conditions, such as using video or virtual reality, these methods fall short of true ecological validity [1].

Recently, rapid development of wearable electroencephalography (EEG) technology offers a promising solution for studying brain activity in naturalistic settings. EEG measures the brain's electrical activity noninvasively with high temporal resolution. It is accessible, affordable, and portable, making it an ideal candidate for mobile neuroimaging applications. These attributes have allowed EEG to support human neuroscience research across diverse domains, including emotional affect, motor functions, social interaction, spatial navigation, and memory [1]. EEG has also been widely applied in consumer and healthcare settings in applications such as sleep monitoring [4], non-invasive brain-computer interface [5], neuromodulation [6], and seizure detection [7]. Recent studies have extensively reviewed EEG's advantages and limitations in real-world contexts [8][9].

One of the biggest challenges in using EEG in real-world settings is ensuring the quality of the signal, as out-of-lab environments present many sources of artifacts [1][10]. Traditional signal processing techniques including digital filtering, frequency decomposition and wavelet-based methods have limited effectiveness due to substantial overlap between brain signal and common artifacts in the frequency domain [11]. Independent component analysis (ICA)-based blind source separation (BSS) is a widely applied standard approach for EEG artifact removal yet requires high dimensional data often not available for wearable EEG, with computation needs preventing real-time application. Methods such as artifact subspace reconstruction (ASR) require careful

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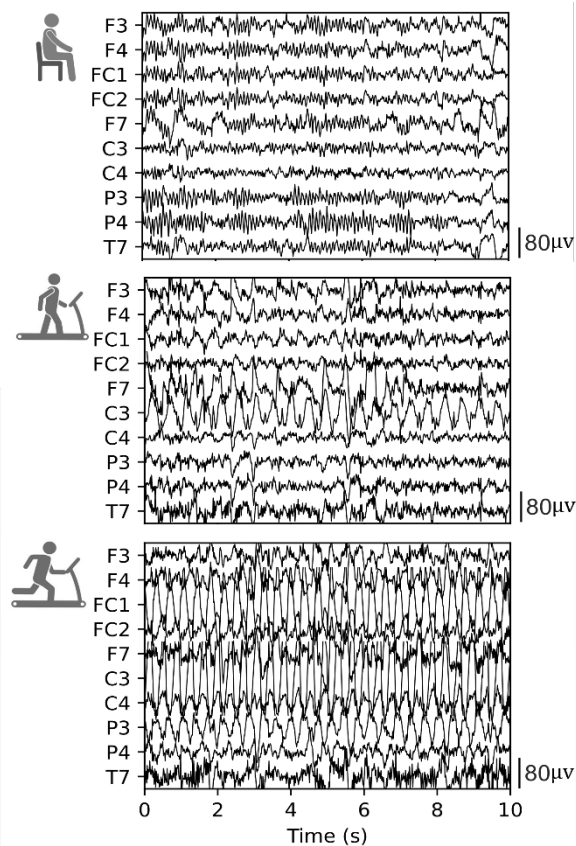


Fig. 1. The experimental paradigm involved data collection during three distinct tasks: rest (with eyes open and closed), walking on a treadmill, and jogging on a treadmill, with each task lasting for 3 minutes. The data corresponding to movement tasks contain high-amplitude, low-frequency fluctuations that correlate with the movement cycle, as well as high-frequency muscle artifacts.

parameter selection [12] that limits efficiency and reproducibility. Additionally, methods such as ASR and deep learning function as “black boxes” with limited explainability, often making it unclear which parts of the data are removed or retained [13], [14]. Some approaches require additional sensors to record from noise sources, which may not fully capture all forms of artifacts while substantially increasing setup complexity and cost [15]. In addition, most artifact removal techniques have limited ability to clean data with very low signal-to-noise ratio, which is often the case in real-world measurements with high amplitude motion artifacts [11]. Developing effective artifact removal techniques that are computationally efficient will improve the feasibility of naturalistic EEG measurements.

As such, we have limited knowledge regarding brain activity during movement, and EEG-based biomarkers have mostly been investigated in resting state. For example, EEG microstate analysis is a promising approach to quantify brain network dynamics, but to our knowledge has not been investigated during walk and jog tasks. EEG microstates are short, quasi-stable periods of brain activity lasting 30-120 milliseconds. They represent discrete and unique spatial patterns of brain activity that are reliable across individuals and across sessions [16][17][18]. Microstate features such as duration, occurrence rate and time coverage has shown promise as reliable

biomarkers for many neurological disorders [19][20][21][22] and have been studied extensively during rest over the past decade.

In this paper, our objective was to develop an efficient and effective method using generalized eigenvalue decomposition (GED) to clean mobile EEG data, then demonstrate the application of EEG microstate analysis on cleaned during-movement data. We first quantified signal-to-noise performance of GED on semi-simulated data – ground truth EEG data with added artifacts derived from simulated or real sources. Subsequently, we compared the method’s performance with common cleaning methods of ASR and ICA. We then applied this method to clean real EEG data collected during tasks involving movement (walking and jogging), evaluating if we can reliably extract microstate maps that are stable across multiple sessions. Lastly, using cleaned data, we compared microstate dynamics across rest and movement tasks for a novel pilot investigation of movement-related EEG microstate changes. This work contributes to the paradigm shift of transitioning human neuroscience studies to the real world.

II. METHODS

A. EEG recording

We collected EEG data from 12 healthy participants (5 females; age = 22.3 ± 2.6 years) using a wireless 32-channel EEG cap equipped with Ag/AgCl active electrodes, arranged according to the 10-20 system on an actiCAP slim cap, and connected to a Brain Products LiveAmp amplifier. Electrodes were filled with SuperVisc conductive gel. We recorded the data with electrode-skin impedances below $25\text{ k}\Omega$, and sampling rate at 500 Hz. FCz electrode was used as the reference. We collected data under four conditions: resting with eyes closed, resting with eyes open, walking on a treadmill and slow jogging on a treadmill. Each activity lasted for 3 minutes. To simulate normal exertion levels for the non-stationary activities, participants each chose comfortable speeds between 2.5 mph to 3.5 mph for walking and between 4.5 to 5.5 mph for running. Fig. 1 presents an example dataset. To assess test-retest reliability, we repeated measurements across four separate sessions for each participant, on two different days and two times of the day (morning and afternoon). Informed consent was obtained from all participants and UBC Research Ethics Board (H20-02313) approved the study.

B. Semi-simulated artefactual EEG data

For validation of our cleaning method, we generated semi-simulated artefactual EEG data. First, we used resting, eyes-open data from session 1 of each participant’s data to extract the ground truth brain activity. The data were initially filtered using a 4th-order Butterworth filter from 1–40 Hz, followed by a notch filter at 60 Hz to remove remaining power line noise. We then re-referenced data to the average of all channels. The MNE toolbox in Python was used for the following processing steps. We used independent component analysis (ICA) with extended infomax algorithm [23], [24] and the ICLabel algorithm [25] to extract only the components labeled as brain (with the brain label having a higher probability than other

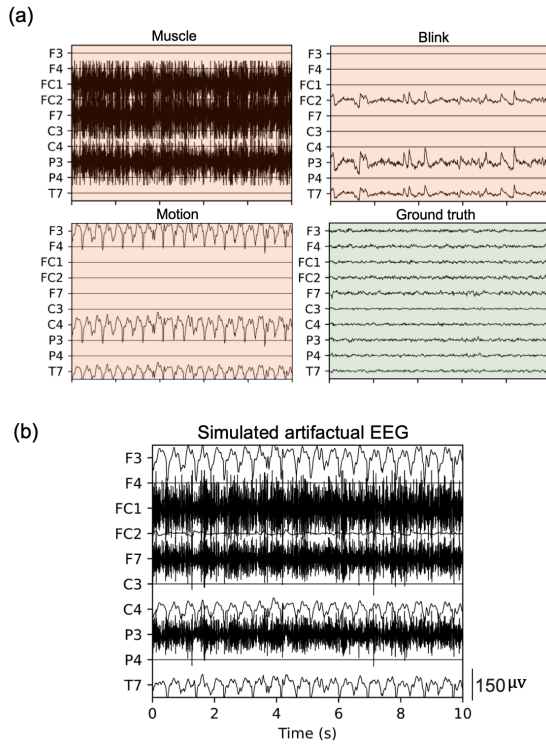


Fig. 2. (a) Shows examples of the individual artifact sources used for simulation. The orange highlighted panels show: EMG artifacts generated using muscle components identified through ICA with filtered white noise; eye-blink artifacts identified through ICA; motion artifacts generated from the rotation of three-axis acceleration of an IMU sensor contained within the amplifier box. The green highlighted panel displays the ground truth EEG data, which are brain components isolated from ICA on resting state eyes-open EEG. (b) presents the resulting semi-simulated artifactual data combining components from panel a.

labels). The classification uses the ICA mixing matrix and power spectral densities of the components as features. Typical brain components show smoothly varying dipolar scalp projections, have power spectra with a $1/f$ pattern, and the power is mainly concentrated between 5 and 30 Hz. In contrast, noise components have different spatial and spectral characteristics from brain components, such as eye movements showing mainly low frequency activity with frontal projections. The reconstructed dataset using only brain components was considered the ground truth brain activity, to which artifacts were added to generate semi-simulated artifactual EEG data. We used two approaches to simulate motion artifacts using the LiveAmp system's inertial measurement unit (IMU) data during a treadmill walking trial. In the first approach, to introduce nonlinearity in mixing the motion artifacts with the ground truth EEG signal, we divided the EEG channels into groups based on their anatomical locations (e.g. frontal, prefrontal, occipital). For each group we generated a random rotation matrix and applied it to the three-axis IMU data. We then identified the axis with the highest variance in the rotated data and added this as a motion artifact component to the ground truth EEG for the corresponding channel group. In the second approach, we used montage information for each EEG

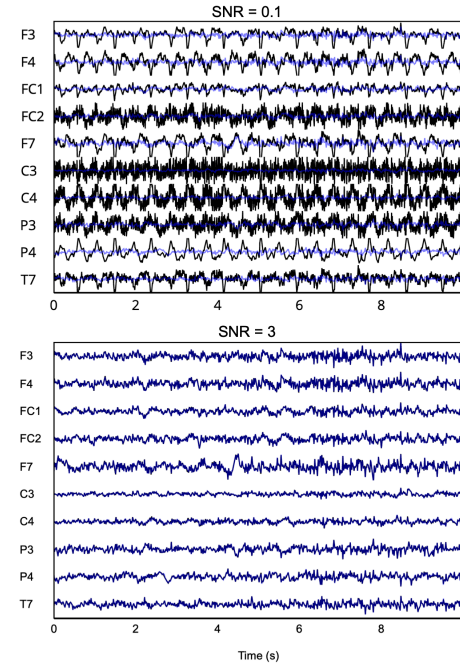


Fig. 3. Semi-simulated EEG data with examples of low and high SNR. The black traces represent the semi-simulated artifactual data, and the blue traces show the underlying clean signal (i.e. ground truth).

channel to extract a unit normal vector at each channel location. We projected three-axis IMU data onto the unit vector corresponding to each channel position, generating 32 unique motion artifact sources in total. These were then added to the ground truth EEG signal. The results of the first simulation are reported in the main manuscript, and the second approach is presented in the Supplementary materials. For simulating EMG artifacts, we used the component labeled as EMG (electromyography) by ICLabel, which was identified in eyes-open data as described earlier in this section. To enhance the representation of EMG artifact under non-stationary conditions, we added white gaussian noise filtered at 20-60 Hz range, following the approach in [26]. To replicate blink/eye movement artifacts, we extracted the component labeled as “eye” by ICLabel from rest eyes open EEG data. EMG and blink artifacts were independently projected into the 32-channel EEG data using a randomly generated projection vector. Fig. 2. provides an illustration of the process of embedding these simulated artifacts into the ground truth brain activity to generate semi-simulated artifactual data. This approach is consistent with previous attempts in the literature to simulate artifactual EEG data [27], [28]. To evaluate the performance of the proposed method under varying signal to noise ratio (SNR) conditions, we define SNR of the semi-simulated data as follows:

$$SNR = \frac{RMS(Ground\ Truth\ data)}{\lambda \cdot RMS(noise)}, \quad (1)$$

Here, RMS refers to the root mean square, calculated as the square root of the sum of squared values of data across all channels and time points. We compute the RMS for both the

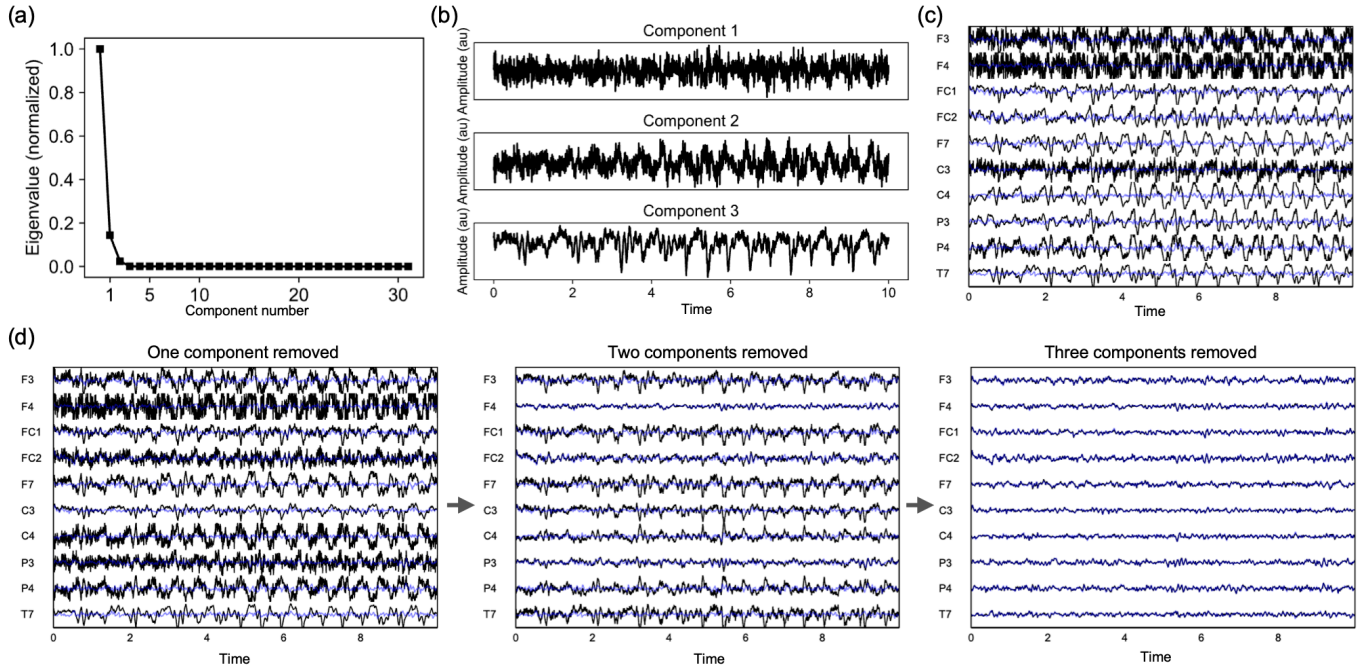


Fig. 4. (a) Normalized eigenvalues extracted from the GED process applied to semi-simulated EEG data (showing example with SNR = 0.1). (b) Time domain amplitudes of the first three components (strongest eigenvalues) isolated by GED. (c) Example EEG channel signals, with the noisy simulated signal shown in black and the ground truth signal in blue. (d) The signals are shown after removing the first, second and the third GED component respectively, demonstrating increasing performance.

ground truth data and the noise data, which includes three types of artifacts: blink, motion, and muscle artifacts. By adjusting the value of λ , we simulated ten varying SNR levels ([0.1, 0.2, 0.3, 0.4, 0.5, 1, 2, 3, 4, 5]), with an emphasis on very low SNRs, as this best replicate the quality of wearable EEG signal collected in real-world scenarios. Fig. 3. presents example semi-simulated signal trace for two different values of SNR.

One participant's data was excluded from this analysis due to missing event marking for the start of the eyes open trial.

C. Contrast based component analysis for artifact removal

Extensive artifacts, such as those caused by movement during data collection, can cause high variability in the data amplitude when compared with controlled conditions like resting-state eyes-closed data. In this methodology, our objective is to perform blind source separation and identify a new subspace where the projection of target data (collected under high-artifact scenarios such as real-world settings) presents high variability, while the projection of background data (controlled resting-state eyes-closed) shows low variability. This approach can be interpreted as finding a viewpoint from which the data's variability highlights artifacts present in the target data but not in the background data.

We used Generalized Eigenvalue Decomposition (GED) to find this subspace where variability is maximized in the target data and minimized in the background data. This approach involves maximizing the ratio of variances, which can be formulated as:

$$\underset{W}{\operatorname{argmax}} \left\{ W^T \left(\frac{\operatorname{Cov}(\text{target})}{\operatorname{Cov}(\text{background})} \right) W \right\} \quad (2)$$

Where $\operatorname{cov}(\text{target})$ is the covariance matrix of artifact-contaminated EEG, $\operatorname{cov}(\text{background})$ is the covariance matrix of relatively clean resting-state EEG, W is the spatial filter being optimized. The optimization finds spatial filters that maximize variance in the target while simultaneously minimizing variance in the background. The resulting eigenvalue associated with each eigen vector quantifies the relative contrast between target and background: a large eigenvalue indicates directions in which the target contains substantially more variance than the background i.e. where artifacts dominate. This property explains why GED can be effective in ultra-low SNR conditions as it identifies directions of maximum variance contrast between the target and the background, in turn enabling the separation and removal of high amplitude noise in these directions. This approach, as explained in [29], does not impose an orthogonality constraint on the components, which is suitable for artifact removal.

The denoising process involves the following steps. First, we compute the covariance matrix of the target (artefactual data we want to denoise). This matrix is then multiplied by the inverse covariance matrix of the background data (resting state data with eyes closed). Next, we extract the eigenvectors of the resulting matrix and sort them in descending order based on their eigenvalues. These eigenvectors define a new coordinate system. We then project the target data onto this coordinate space and visualize the components. The components corresponding to the largest eigenvalues, where artifacts are most prominent, are removed to denoise the data. The number of components to remove is the only hyperparameter of this

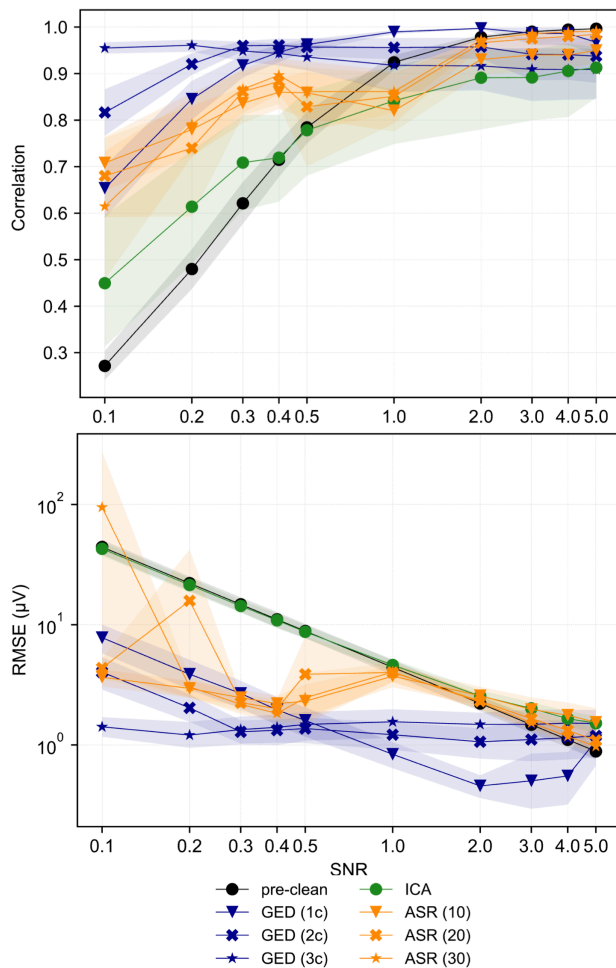


Fig. 5. Comparison of artifact removal performance across SNR levels. Top: correlation between cleaned and ground truth EEG signals. Bottom: RMSE between cleaned and ground truth signals. Results are shown for GED (removing the first 1, 2, or 3 components: 1c, 2c, 3c), ICA (Infomax with ICLabel brain components retained), and ASR with cutoff values of 10, 20, and 30. Shaded regions indicate 95% confidence interval.

method and would be determined by the researcher. For semi-simulated data, we investigated removing 1, 2 and 3 components. For real data, we identified removal of first 3 components to be optimal based on the increase in the ratio of brain components (Supplementary materials), followed by visual inspection to confirm the artifactual patterns in these components.

D. Evaluation of the GED methodology on semi-simulated data

To assess the effectiveness of the GED method on the semi-simulated data, we used two evaluation metrics. First, the Pearson's correlation between the ground truth signal and the cleaned data was calculated for each channel, and the results were averaged across all channels to calculate a single correlation value. Second, the root mean squared error (RMSE) was computed using all data points from the ground truth and preprocessed signals. We evaluated the method across different

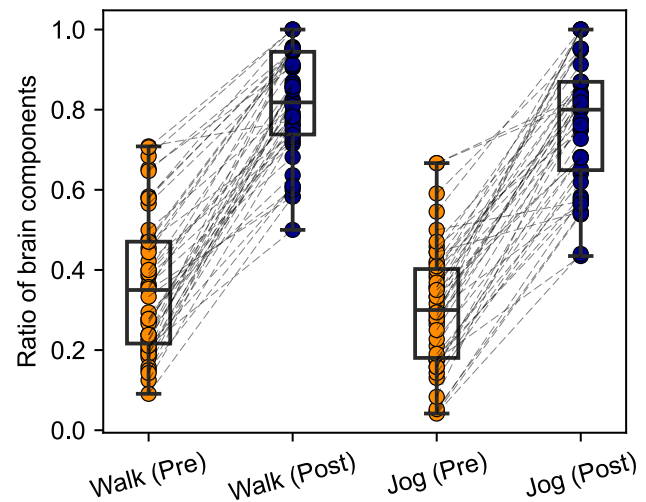


Fig. 6. Ratio of brain components to total components identified by ICLabel for both walking and jogging tasks before and after applying GED. Individual data points are presented and dashed lines connect the pre-GED and post-GED ratios for the same subjects. The results show an increase in the ratio of brain components after applying GED for the majority of subjects in both tasks.

levels of SNR to assess its performance under varying noise conditions. Additionally, we compared the method with two well-established methods in the literature: ICA and ASR. For ICA, as with the resting state data, we used the Infomax algorithm [23] and retained only the components labeled as “brain” by ICLabel [30]. For ASR, we used standard default parameters, and evaluated performance at three cutoff values of 10, 20 and 30 as suggested to be within the optimal range by multiple studies [31], [32]. As ASR also requires a baseline clean section of data, we concatenated the resting data with simulated data before applying the method.

E. Evaluation of the GED methodology on real data

Since ground truth is not available for real data, we aimed to assess the improvement in signal quality using the widely applied method of ICA and ICLabel. Specifically, we evaluated the number of “brain” components identified before and after applying the GED method on walking and jogging data, using the Infomax ICA algorithm [23], [24] and the ICLabel algorithm [25]. In this evaluation, an increase in the number of “brain” components, as labeled by the ICLabel algorithm, indicates a potential improvement in signal quality after applying the GED algorithm. If more dipolar brain ICs are extracted, then it means that denoising helped ICA separate the data into meaningful neural sources rather than contaminated components. It is important to note that because the GED process reduces the dimensionality of the data, we reported the ratio of “brain” components to the total number of components, as identified by the ICA algorithm in MNE. This approach provides a more comparable metric between pre- and post-GED data. This analysis was performed across data from all subjects and for all four sessions, with the exception of one walking session that was excluded due to mislabeling.

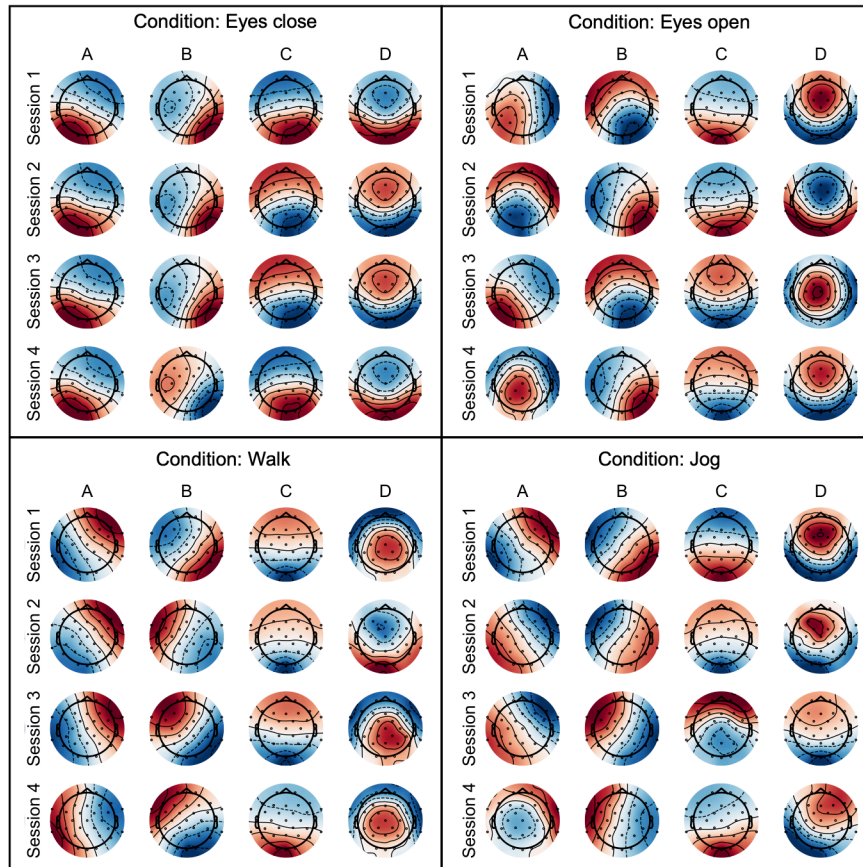


Fig. 7. Microstate maps extracted separately for each task and session using data from all subjects.

F. Microstate analysis

We used the Pycrostates package in Python for microstate analysis [33]. Global Field Potential (GFP) peaks were extracted from individual datasets using built-in functions. We then calculated GFP as the standard deviation of all channel values at each time point resulting in a single GFP time series. Peaks of GFP were identified with a minimum distance of 2 samples between peaks. To ensure consistency and avoid bias from differences in the number of peaks across subjects, we identified the minimum number of GFP peaks across all datasets and resampled each subject's GFP series to match this minimum. The GFP peaks were clustered using a modified k-means algorithm, following methods commonly used in microstate research. We extracted 4 canonical microstates (A-D) as described in previous microstate research [16], [17], [34], [35], [36]. The data from each individual subject were then backfitted to the identified microstates. We applied a smoothing parameter of 12 samples (24 ms) to reduce rapid fluctuations caused by noise or artifacts. Polarity was not considered in this process consistent with the literature on characterizing microstates.

To assess consistency, we calculated the average correlation between microstate maps to evaluate the similarity of the topographical patterns of each microstate across sessions for each task. Additionally, we evaluated Global Explained Variance (GEV) using all four microstates backfitted to individual subject data and reported the value for each session

and each subject separately to assess the quality of the model in explaining the data for all the tasks. Lastly, we compared the extracted microstates of the first session for each task with the microstate templates that has been previously reported in the literature (Supplementary figure 6). As a proof of concept, we also evaluated differences in microstates' duration, occurrence and time coverage across tasks using data recorded in session 1. To perform this analysis, we first generated a common set of microstates using the combined data from all participants across the entire session (including all four tasks). These group-level microstates were then backfitted to each individual's data to extract the three microstate dynamics measures. We used repeated-measures analysis of variance (ANOVA) with a significance threshold of 0.01 to account for multiple comparisons. When significance was found, we conducted post hoc analyses using multiple paired t-tests with Bonferroni-corrected threshold of 0.008 to identify specific task-level differences.

III. RESULTS

A. Evaluation of GED performance as a novel preprocessing method on semi-simulated data

Fig. 4 demonstrates an example case of how the method improves signal quality by removing three components modeling the original artifact sources in semi-simulated data with low SNR. Fig. 5 shows aggregated results from different participants, indicating GED performs better in low SNR



Fig. 8. (a) Heatmap showing the average correlation between microstate maps across all sessions illustrating the consistency of maps across all sessions within each task. Correlations were computed separately for each task. (b) Figure presents the GEV value for each subject and each session illustrating high explained variance using four microstates across all tasks.

conditions when removing 3 component and higher SNR conditions when removing only the first component. The average post-cleaning ground truth correlation is 0.93 and RMSE is 1.43. For low-SNR values (less than 0.5), removing three components (the first three extracted by GED) performs better, whereas for higher-quality signal (with higher SNR), removing just one component is sufficient to restore the ground truth data.

As shown in figure 5, GED outperformed ICA and ASR across a range of SNRs. In the correlation analysis, GED achieved higher correlation with the clean signal particularly at low SNRs (<1) where ASR and ICA showed reduced performance. Similarly, in terms of RMSE, GED produced lower errors, indicating more accurate recovery of the ground truth brain data.

B. Evaluation of GED performance on real data

The ratio of "brain" components to the total number of components identified by the ICLabel algorithm in MNE is presented in Fig. 6. This ratio was lower before applying GED, with an increase observed post-GED for the majority of subjects. On average, the number of brain components increased by 10.9 and 11.8 for the Walk and Jog tasks, respectively.

C. Microstate Analysis

Fig. 7 presents the maps extracted from all the subject's data across sessions for each task separately. Microstate A is labeled based on right frontal to left posterior scalp topography. Microstate B is labeled based on left frontal to right posterior topographical map. Microstate C is characterized by symmetric anterior to posterior scalp configuration. Lastly, microstate D characterized by fronto-central scalp map configuration. A detailed description of these canonical maps is provided in [17].

In the resting-state eyes closed condition, all four canonical microstates were consistently extracted across sessions, and their topographies aligned with prior microstate literature. In other tasks, showed less consistency as illustrated in figure 8 (a). In the walking condition, the topographical map of the last microstate in the final two sessions resembled what has been labeled as microstate E in [17]. However, several studies have report a similar topography as microstate D highlighting inconsistencies in labeling conventions [37]. To further assess similarity to canonical maps, we compared the microstate topographies with six previous studies [38], [39], [40], [41], [42], [43] in Supplementary figure 8. The correlations were consistently high across most conditions (typically >0.8).

To assess how well the extracted microstates represent the data, the global explained variance (GEV) values are reported in Fig. 8(b). High GEV values were observed across all individual subjects and sessions, confirming that the extracted microstates capture the underlying brain dynamics. The average GEV values for each task across all subjects and sessions were calculated as: Eyes closed: $74 \pm 6\%$, Eyes open: $69 \pm 4\%$, Walk: $63 \pm 6\%$, and Jog: 61 ± 8 . The figure also demonstrates that the microstates are not biased toward specific subjects, as GEV values are not widely variant across individuals. Additionally, GEV remained consistent across sessions and was also high for motion tasks, which further confirms that well-established microstates explain the data even in motion conditions with post-GED data.

Lastly, Fig. 9 presents the global microstate maps generated using data from session 1. We observed significant task-related differences in microstate duration, occurrence, and time coverage. Specifically, microstate A showed task-dependent modulations across all three features. Microstate B showed significant effects in duration only. Microstate D displayed significant modulations in occurrence rate and time coverage. The statistically significant findings are summarized in Table I and visualized in Fig. 9.

IV. DISCUSSION

Traditional neuroimaging studies are typically restricted to stationary conditions or highly controlled tasks, limiting our understanding of brain function during natural and dynamic behaviors. Yet, studying brain dynamics during real-world motion remains challenging due to movement related artifacts. In this work, we address this challenge by introducing and validating a novel artifact removal method based on generalized eigen decomposition using mobile EEG data recorded during walking and jogging. We demonstrate the method's

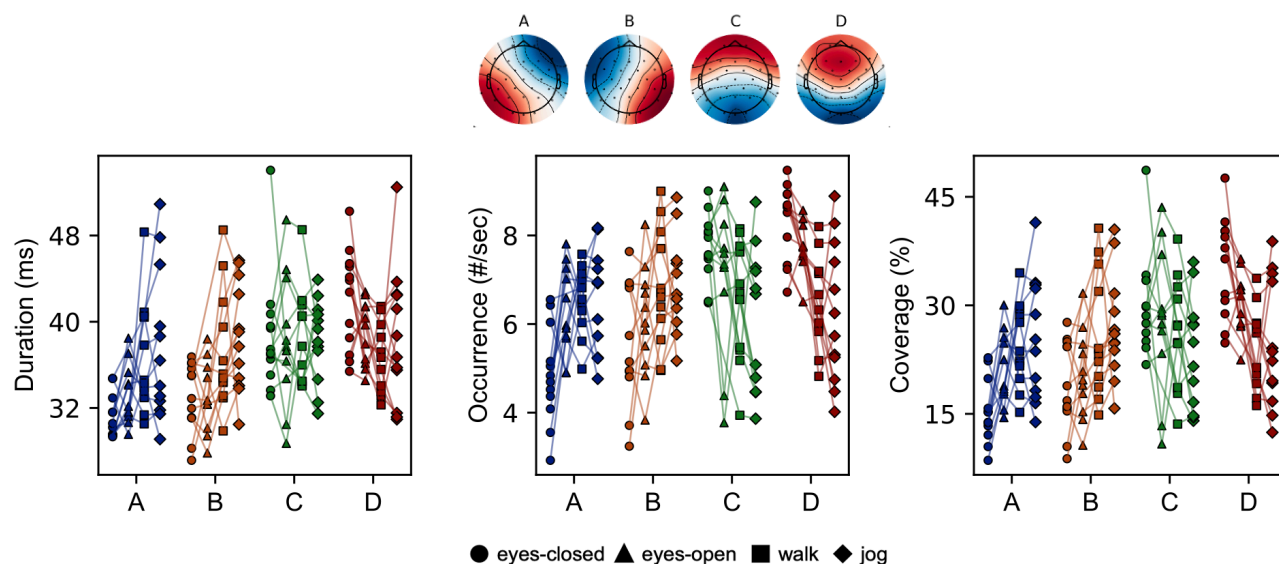


Fig. 9. Microstate dynamics across four tasks (eyes closed, eyes open, walk, jog) for microstates A–D. Top row shows the topographical maps of the four global microstates extracted from session 1. Bottom row displays subject-level comparisons of microstate duration (left), occurrence rate (middle), and time coverage (right). Each marker shape represents a different task (circle: eyes closed, triangle: eyes open, square: walk, diamond: jog). Lines connect data points from the same subject across tasks. Horizontal bars indicate significant task-related differences (repeated-measures ANOVA, followed by post hoc comparisons with Bonferroni correction, $p < 0.008$).

effectiveness on semi-simulated data, showing successful recovery of ground truth signals in very low SNR conditions.

While ground truth is not available for real data, previous studies have used the increase in brain-related components identified by ICA as an indirect measure of data quality [44], [45]. In our evaluation of the real data, we similarly observed consistent increases in the proportion of brain-related components following artifact removal. To further validate the method, we tested the feasibility of extracting canonical microstates during motion. The results showed that microstates were consistently preserved, showing stable spatial topographies and high global explained variance across sessions. Although microstate analysis is an indirect validation approach, it can serve as a powerful indicator of data quality specifically when artifacts have high amplitude characteristics such as motion. This is because microstate analysis operates on the local maxima of the GFP series where time points with the absolute voltage across channels is high. If the data contain high amplitude noise with topographies that deviate from typical clean EEG patterns, those topographies will be included in GFP peaks and influence the clustering. As a result, k-means clustering may form spurious microstate classes that do not resemble canonical patterns, highlighting the presence of residual artifacts. Therefore, the successful recovery of typical microstates under motion provides strong indirect evidence for artifact attenuation.

Finally, we use the cleaned data to demonstrate a novel analysis of brain microstates during movement, where we identified significant task related differences in microstate

dynamics between rest and motion tasks. This line of investigation is particularly important because the functional significance of microstates remains an open discussion. By examining how microstate characteristics vary across tasks, we gain insights into how task modulation influences the dynamics of EEG trajectories and contribute to a broader understanding of large-scale brain dynamics and functional role of microstates in behavior.

In this work, we demonstrate a novel application of contrastive based component analysis method for removing high amplitude movement related artifacts from EEG data. Linear contrast-based component analysis methods have been introduced in prior work for different applications. For example, [46] introduced contrastive PCA with the idea that the pattern of interest in the data might not necessarily correspond to the largest source of variability, but can be informed by the relative variability when contrasted with a related background dataset. Similarly, the GED method identifies contrastive components with the orthogonality constraint removed. In a related application, [29] used GED to isolate neuronal population activity specific to a particular task while disentangling it from background brain activity. Interestingly, GED has also been applied to mitigate power line noise artifacts [47]. In this case, a narrowband signal serves as the background activity, while a broadband signal is used as the target activity, to reduce signal energy within a specific frequency range. GED's application as part of the spatial filter family has been extensively explored in previous studies for different

TABLE I
TASK RELATED CHANGES IN MICROSTATE DYNAMICS

Feature (N = 11)	Microstate	F (3,30) [p-value]	Task Comparison [95% CI mean difference]	p-value (Cohen's d)
Duration	A	5.87 [p = 0.003]	Eyes Closed < Walk [3.4, 9.7]	0.007 (1.55)
			Eyes Closed < Jog [4.1, 12.1]	0.005 (1.59)
	B	5.56 [p = 0.004]	Eyes Closed < Jog [3.3, 8.8]	0.004 (1.23)
Occurrence	A	10.45 [p < 0.001]	Eyes Closed < Eyes Open [0.6, 1.1]	0.007 (1.60)
			Eyes Closed < Walk [1.4, 2.7]	<0.001 (2.07)
			Eyes Closed < Jog [1.4, 2.8]	<0.001 (1.89)
	D	16.02 [p < 0.001]	Eyes Closed > Walk [1.5, 2.8]	< 0.001 (2.12)
			Eyes Closed > Jog [1.6, 3.2]	< 0.001 (1.90)
			Eyes Open > Walk [1.2, 2.4]	0.003 (1.59)
			Eyes Open > Jog [1.2, 2.8]	0.005 (1.48)
Coverage	A	8.33 [p < 0.001]	Eyes Closed < Walk [6.9, 14.3]	<0.001 (1.94)
			Eyes Closed < Jog [7.3, 16.1]	<0.001 (1.84)
	D	13.91 [p < 0.001]	Eyes Closed > Walk [7.5, 16.8]	< 0.001 (1.84)
			Eyes Closed > Jog [9.0, 17.3]	< 0.001 (1.71)
			Eyes Open > Walk [5.6, 12.5]	0.007 (1.22)

applications [48], [49], [50], [51]. Our results demonstrate that GED is also highly effective in addressing EEG artifacts from multiple sources during mobile recording.

A major challenge in collecting EEG during naturalistic movement is that motion-related artifacts often have unknown frequency range and signal amplitudes several orders of magnitude larger than the underlying neural signals, resulting in extremely low SNRs. Most existing EEG denoising methods have been validated under moderate artifact conditions but their performance deteriorates in ultra-low SNR regimes. For instance, the EEMD-CCA method, validated on semi-simulated data contaminated with EMG artifacts, show performance degradation as SNR drops below 1 [26]. Similarly deep learning based methods have primarily been evaluated in low to moderate SNR regimes (e.g. SNR>0.5) (e.g. [28][52]). Our two approaches of nonlinear motion artifact simulations produced highly contaminated datasets spanning a wide range of SNRs, including ultra-low regimes (<0.5) that more realistically reflect real walking and jogging EEG (Fig. S5). When artifacts were severe (SNR < 0.5), removing the first three GED components resulted in the best recovery. Beyond subject-specific cleaning, we further showed that GED maintains strong performance even when background covariance data are derived from other participants (Fig. S7). This suggests that the method can still perform well even when not calibrated for individual participants similar to approach in [53], indicating generalizability of the structural properties of clean EEG data. However, because the GED approach relied on the spatial covariance structure of the EEG signals, the background dataset needs to have matched electrode layout and referencing scheme which could be achieved through a high-density standard background dataset that enables selection of different montages for each application. Another advantage of this contrast-based linear approach is its flexibility in real-time application. Given sufficient calibration or background data, the method imposes

no strict constraints on the temporal quantity of the data needed for artifact removal, which provides the opportunity for real-time applications where short windows of contaminated EEG can be cleaned in real time. This can have applications for real-time mobile neuroimaging or brain-computer interface frameworks. Additionally, the robustness to motion artifact removal makes it particularly suitable for monitoring brain activity during movement-based tasks or in populations with motor symptoms, such as individuals with Parkinson's disease.

We validated the proposed denoising method using both semi-simulated and real EEG data across a wide range of SNR conditions. As a secondary validation, we leveraged microstate identification as an approach to interpret neural dynamics post denoising. All microstate topographies extracted closely aligned with previous literature [16], [17] and supplementary figure 8. For the walk condition, microstates A, B and C showed stable topographies similar to rest conditions. The fourth microstate in this condition had topography resembling microstate E [17]. The consistency of microstate maps across sessions suggests strong test-retest reliability, thus effective cleaning of data, as motion-related artifacts distort the signal and can lead to formation of clusters that are not well defined and potentially spurious microstate maps. For the jog condition, microstate maps were extracted, however, the resulting maps were less consistent with canonical microstate topographies reported in the literature compared to the other tasks. This discrepancy may be due to residual artifacts that remained in the data that distorted the cluster centroid and consequently the resulting microstate maps.

As a novel demonstration of brain microstate analysis using cleaned data, we observed significant differences in microstate measures across tasks, particularly during motion. While microstates are well-studied in resting-state and stationary cognitive paradigms, their dynamics during physical activity remain unexplored. Insights from EEG-fMRI and EEG source localization studies suggest that specific microstates correspond

to activity in distinct functional brain networks [54], [55], providing a potential framework for interpreting our findings in the context of naturalistic behavior. The high GEV values observed for both rest and motion conditions support the application of microstate analysis in modeling whole brain dynamics. Microstate A, characterized by a right frontal to left posterior topography, has been associated with the auditory network and temporal cortex activity [54], [55]. It has been reported to show increased presence during verbalization and visualization tasks, although these findings are not always consistent (see review in [17]). Some studies have also linked longer durations of microstate A to heightened alertness and arousability [56][38]. In our study, its increase presence during walking and jogging may reflect elevated arousal or enhanced sensory processing.

Microstate B, with a left frontal to right posterior configuration, has been linked to visual processing areas and is consistently reported to increase in response to visual stimuli and tasks involving visuospatial attention [38][55][41]. We observed increased duration of this microstate during jogging tasks. While three of four tasks involved open eyes, significant changes were only observed when comparing rest to jogging condition which may reflect more demand in visuospatial processing during jogging.

Microstate D is defined with fronto-central configuration. Its reduced presence during movement may reflect both methodological limitations and functional differences. Previous studies have reported an increase in this microstate during cognitively demanding tasks [57][40][41]. The reduced duration, occurrence, and coverage of microstate D during motion tasks might reflect limited cognitive resource availability when participants are engaged in physical activity.

Overall, our preliminary findings suggest that microstate dynamics, hypothesized to reflect large-scale brain network activity are modulated during tasks involving movement. Future studies should systematically investigate these dynamics using larger datasets and more refined task designs to better understand how brain networks dynamics operate in ecologically valid settings.

This study has several limitations. First, the cleaning method is validated using both semi-simulated and real EEG data collected with a research-grade wireless EEG system. Additional validation on data from lower-cost, lower-quality wearable systems was not in the scope of the current paper but will be important to conduct for broader applicability. Secondly, GED requires a subjective decision about the number of components to remove. This can be adjusted based on the dataset and future studies can refine the criteria of removing components by generalizing the methodology on larger datasets. Thirdly, while simulated artifacts incorporated nonlinear transformations of the motion signal, the GED method is based on linear decomposition and thus may not fully model nonlinear source interactions in real EEG data. It should also be noted that the validation of the methodology on real data was performed using ICLabel algorithm, which was trained using resting state data and lacks a distinct "motion" artifact category. Lastly, while our interpretations of microstate functional relevance are informed by prior studies, linking these topographies to specific brain networks or behavioral patterns remains an open question and future studies should incorporate

higher density EEG enabling source localization to confirm the corresponding network-level mechanisms.

V. CONCLUSION

Overall, this study contributes to the literature by demonstrating the effectiveness of GED for denoising EEG data contaminated with high-amplitude artifacts even when these artifacts overlap with relevant frequency bands. Using semi-simulated data containing eye movement, EMG and multiple sources of motion artifacts, we showed that GED outperforms established methods such as ASR and ICA in very low SNR regimes.

This work also highlights that reliable brain microstate extraction is feasible across a range of tasks involving physical movement such as walking and jogging, supporting the use of EEG to assess brain state dynamics in naturalistic, high motion scenarios. This work support ongoing efforts to study brain activity in ecologically valid scenarios.

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